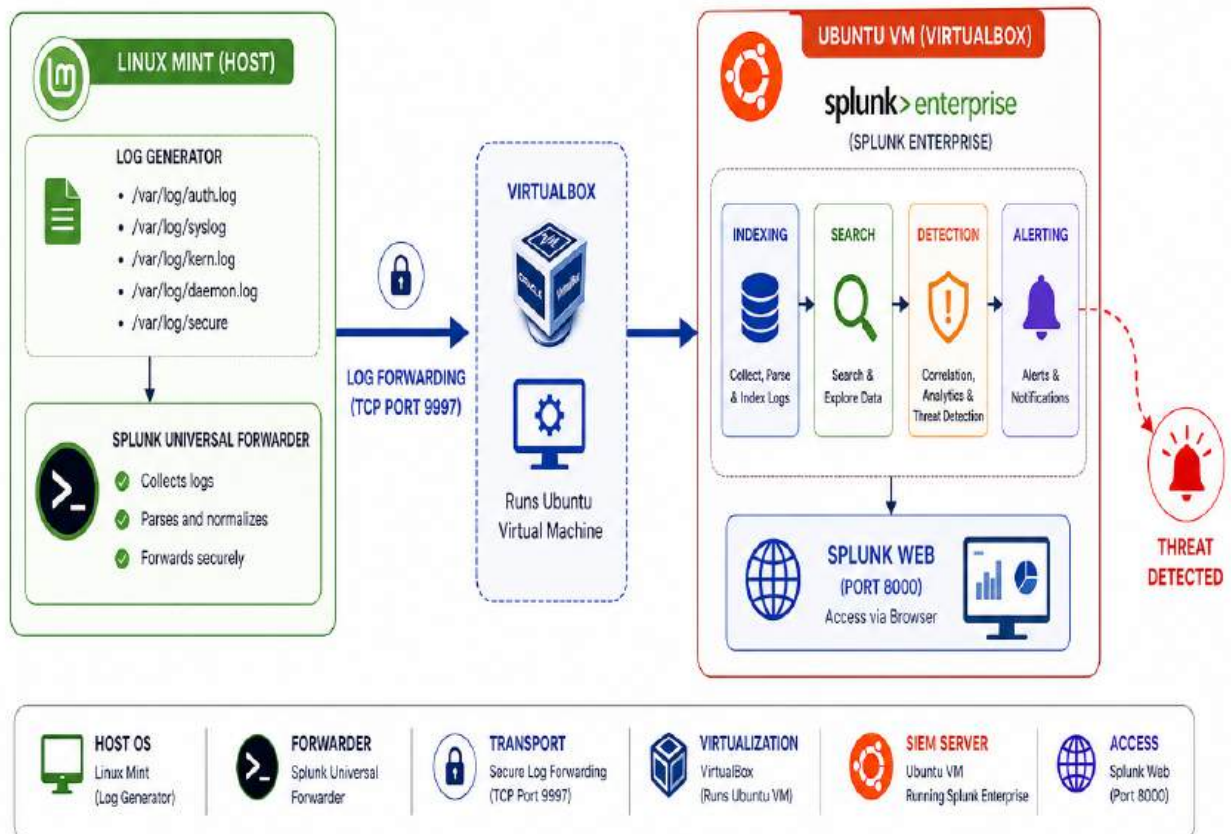


END-TO-END SOC LAB ARCHITECTURE USING **SPLUNK** ENTERPRISE



Collect. Analyze. Detect. Alert.



LOGS TODAY. **INSIGHTS TOMORROW.** SECURITY ALWAYS.



End-to-End SOC Lab Implementation using Splunk Enterprise

Executive Summary

This project presents the design and implementation of an end-to-end Security Operations Center (SOC) lab using Splunk Enterprise and Universal Forwarder.

The objective of this lab is to simulate a real-world SOC environment where logs are collected, forwarded, monitored, and analyzed to detect suspicious activities and potential security threats.

A Linux Mint system was used as the log-generating machine with the Universal Forwarder installed, while an Ubuntu-based virtual machine hosted the Splunk server for centralized log collection and analysis.

The project demonstrates the complete log pipeline, including forwarder configuration, data ingestion, attack simulation (SSH brute-force), detection using search queries, and alert creation for suspicious events. Troubleshooting techniques were also applied to resolve real-world issues, and SSL-based secure log transmission was explored.

This lab provides practical, hands-on experience in key SOC operations such as log management, threat detection, alerting, and incident analysis, forming a strong foundation for a SOC analyst role.

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1. Introduction

A Security Operations Center (SOC) is responsible for continuously monitoring and analyzing security events to protect systems from cyber threats. Log monitoring plays a critical role in identifying suspicious activities such as unauthorized access, brute-force attacks, and system anomalies.

In this project, an end-to-end SOC lab is implemented using Splunk Enterprise and Universal Forwarder. The objective is to simulate a real-world SOC environment where logs are collected from a host machine and forwarded to a centralized monitoring system.

This helps in understanding how security analysts detect and respond to threats, analyze logs, and create alerts for suspicious activities. It also provides hands-on experience with log ingestion, monitoring, and basic threat detection techniques.

The main goal of this project is to build a practical SOC environment and gain foundational skills required for a SOC analyst role.

2. Lab Setup

The lab environment for this project is created using a host machine and a virtualized setup to simulate a real-world SOC architecture.

The host system is running Linux Mint, where the Splunk Universal Forwarder is installed. This forwarder is responsible for collecting system logs and sending them to the central server.

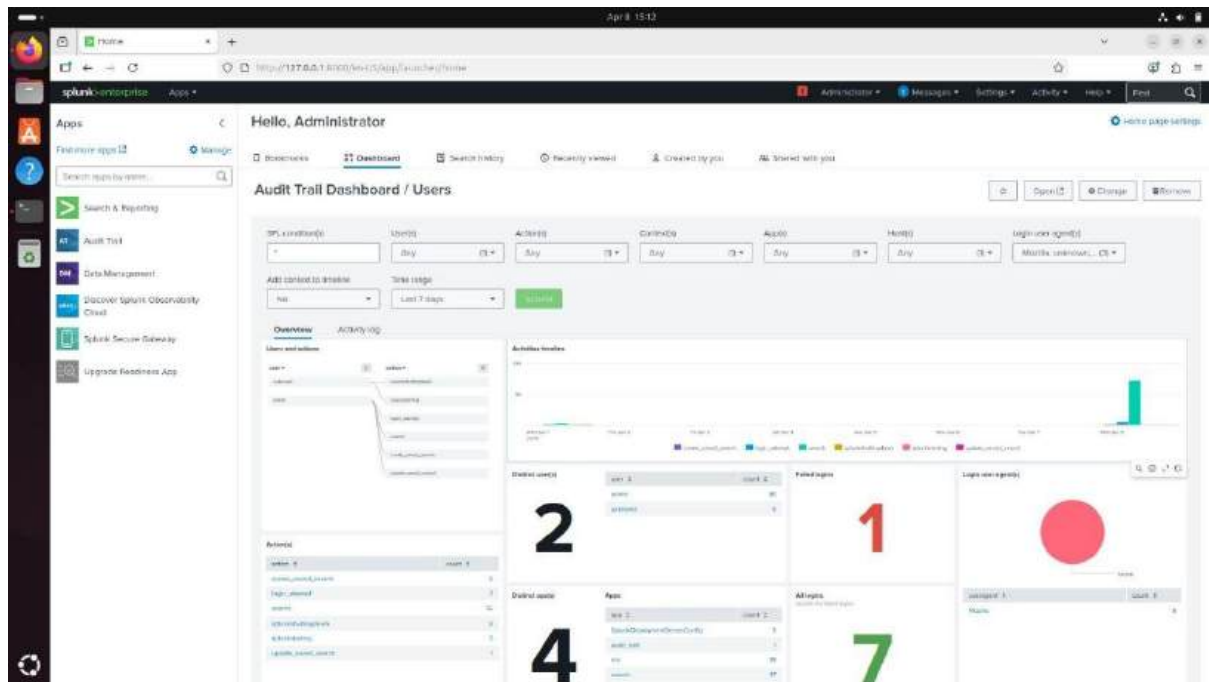
A virtual machine is created using VirtualBox, where Splunk Enterprise is installed on an Ubuntu-based system. This acts as the central log monitoring server that receives and analyzes logs from the forwarder.

The lab setup includes the following components:

- Host Machine: Linux Mint (Log generator + Universal Forwarder)
- Virtual Machine: Ubuntu (Splunk Enterprise Server)
- Virtualization Tool: VirtualBox
- Log Forwarding Tool: Splunk Universal Forwarder

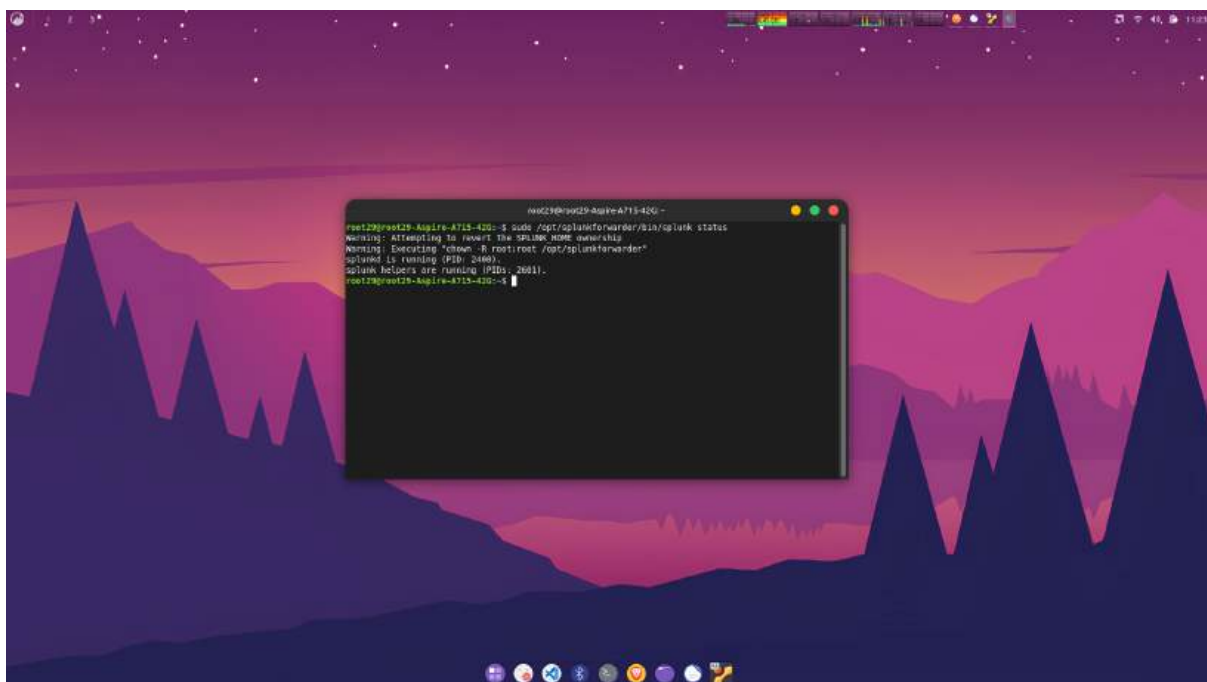
This setup allows logs generated on the host machine to be securely forwarded to the Splunk server for monitoring and analysis, simulating a basic SOC environment.

Splunk Dashboard:-



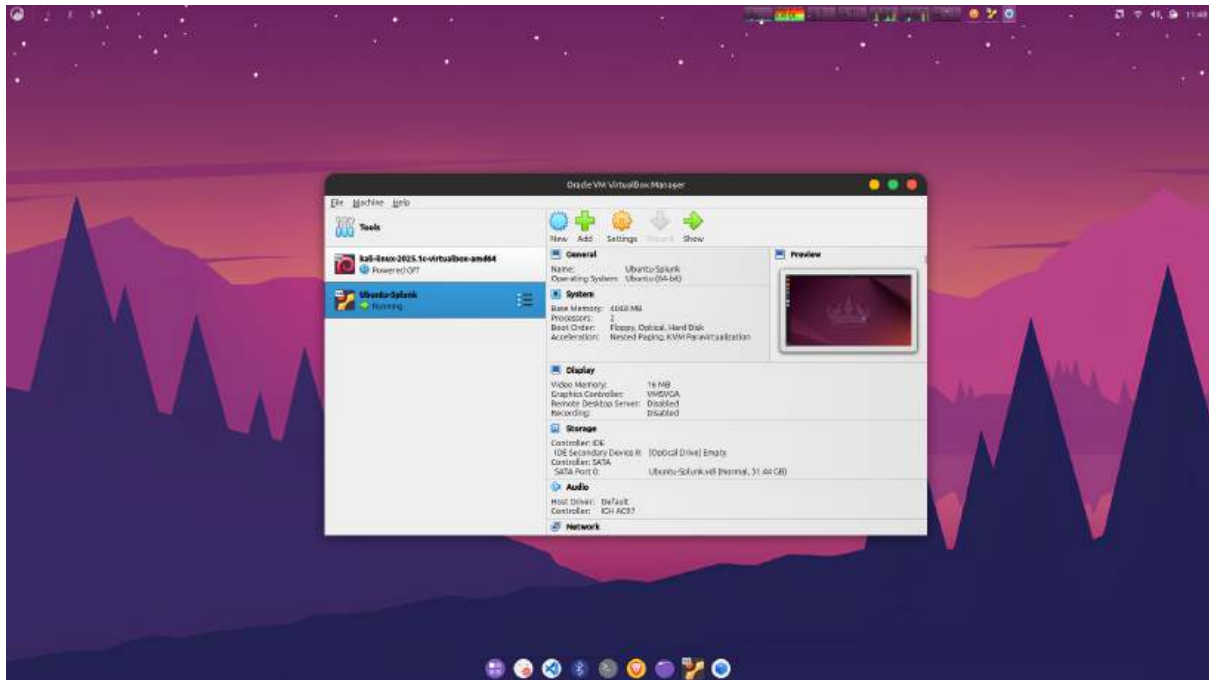
"This screenshot shows the Splunk Enterprise dashboard running on the virtual machine."

UF Running Terminal:-



“This screenshot confirms that the Splunk Universal Forwarder is successfully running on the host system.”

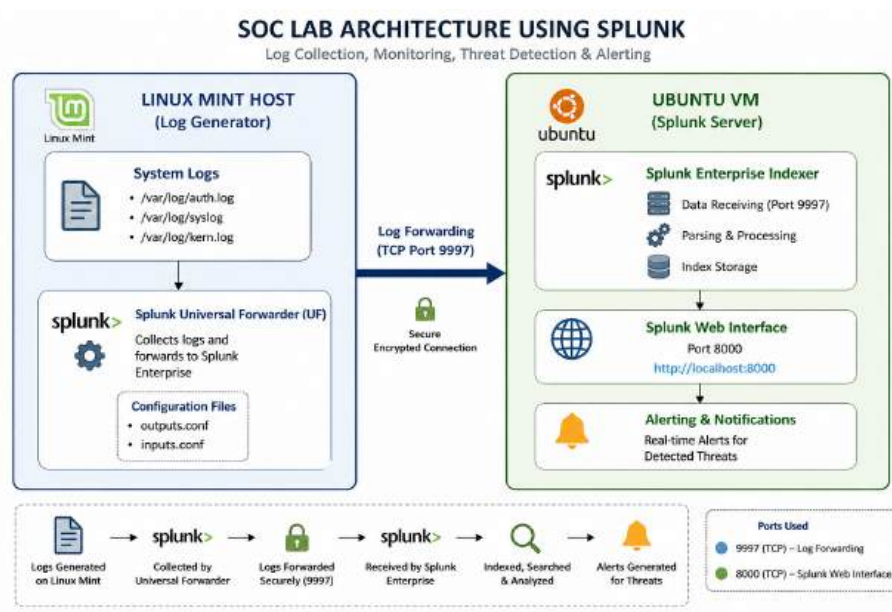
VirtualBox:-



“This screenshot shows the virtual machine environment created using VirtualBox for running the Splunk server.”

2.1 System Architecture Diagram

End-to-End SOC Lab Architecture using Splunk Enterprise and Universal Forwarder



3. Configuration

In this step, the Splunk server and Universal Forwarder are configured to enable log forwarding from the host system to the Splunk server.

3.1 Enabling Data Receiving on Splunk

To allow the Splunk server to receive logs from the forwarder, data receiving is enabled on port 9997.

Steps:

- Navigate to Settings → Forwarding and Receiving
- Click on “Configure Receiving”
- Add a new port (9997)

This allows the Splunk instance to listen for incoming log data from forwarders.

3.2 Configuring Universal Forwarder

The Universal Forwarder is configured to send logs to the Splunk server by editing the configuration files.

The outputs.conf file is used to define the Splunk server IP and port:

Example configuration:

```
[tcpout]
```

```
defaultGroup = default-autolb-group
```

```
[tcpout:default-autolb-group]
```

```
server = <Splunk_Server_IP>:9997
```

The inputs.conf file is used to specify which logs should be monitored:

Example configuration:

```
[monitor:///var/log/auth.log]
```

```
disabled = false
```

This configuration ensures that authentication logs are monitored and forwarded to the Splunk server.

3.3 Verifying Network Connectivity

To ensure proper communication between the host and virtual machine, network connectivity is tested using the ping command.

Example:

```
ping <Splunk_Server_IP>
```

Successful responses confirm that both systems can communicate with each other.

3.4 Restarting the Forwarder

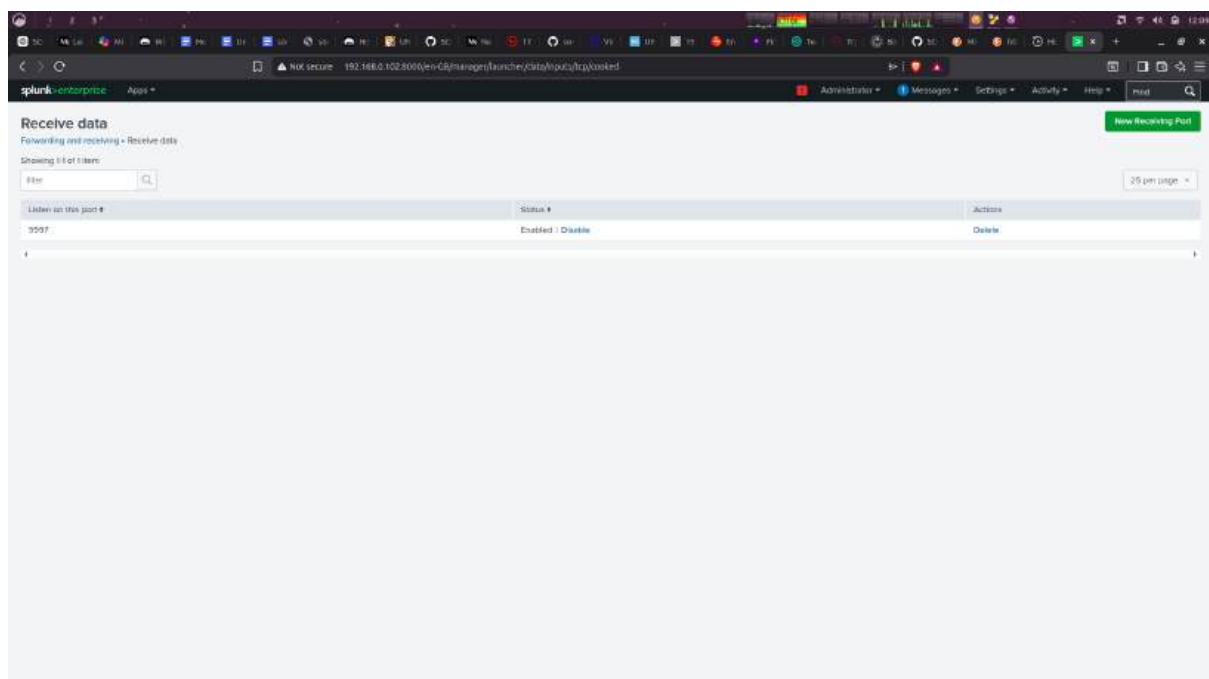
After making configuration changes, the Universal Forwarder is restarted to apply the settings.

Command:

```
sudo /opt/splunkforwarder/bin/splunk restart
```

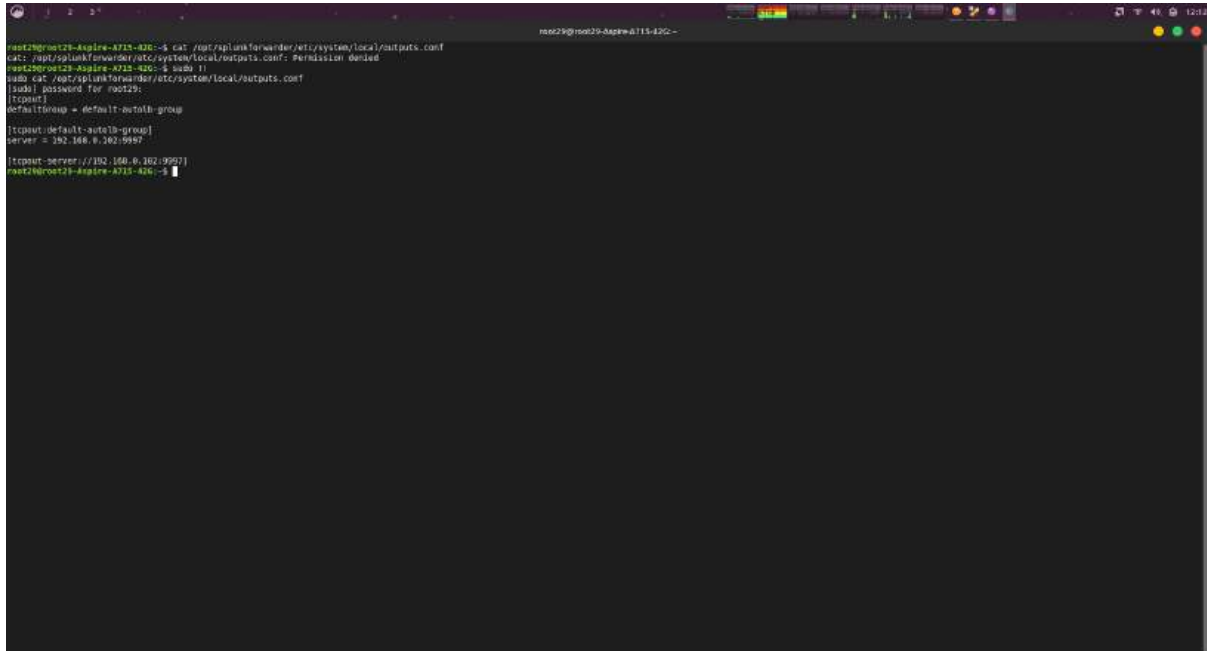
This ensures that the forwarder starts sending logs to the Splunk server.

A. Splunk Receiving (Port 9997)



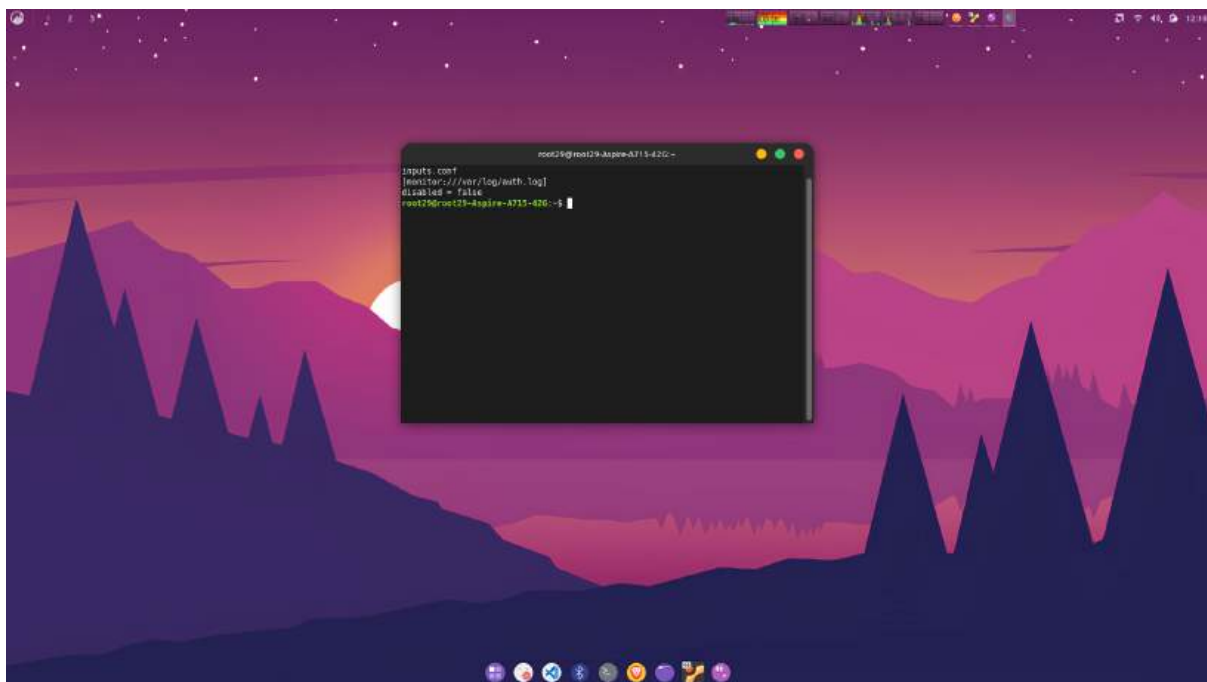
B. Universal Forwarder Configuration (outputs.conf & inputs.conf)

outputs.conf: - Configured the forwarder to send logs to the Splunk server.

A terminal window with a dark background. The prompt is 'root29@root29-Aspire-A715-42G:~'. The user enters 'cat /etc/splunkforwarder/etc/system/local/outputs.conf'. The output shows the configuration for the tcpout stanza, including the server IP and port.

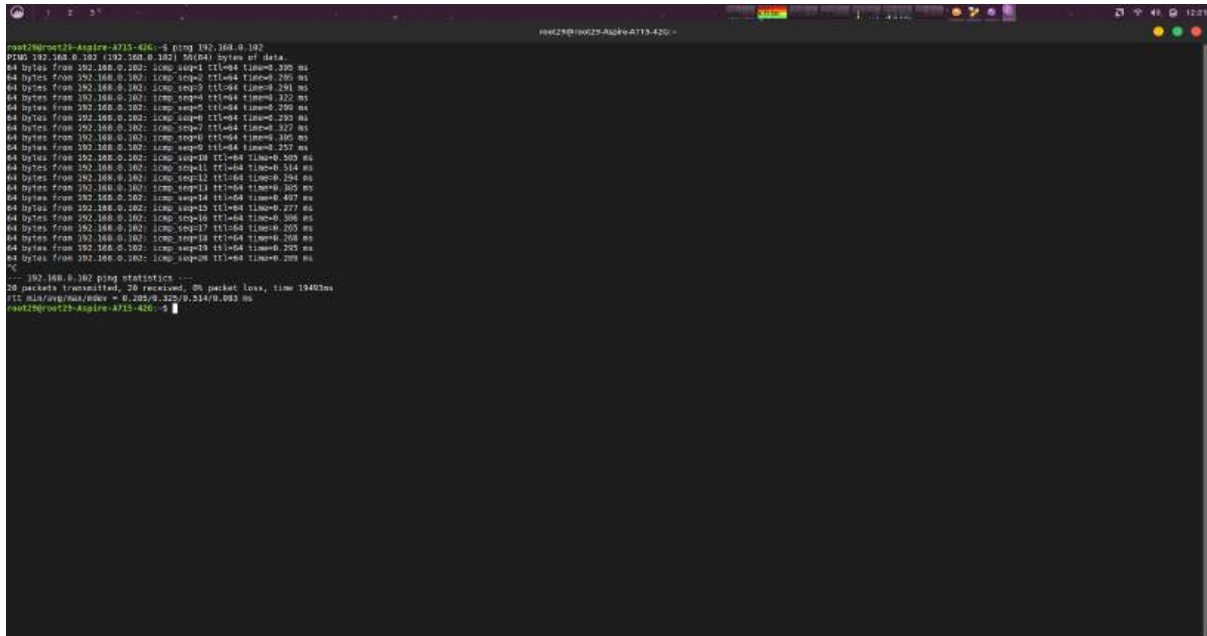
```
root29@root29-Aspire-A715-42G:~$ cat /etc/splunkforwarder/etc/system/local/outputs.conf
cat: /etc/splunkforwarder/etc/system/local/outputs.conf: Permission denied
root29@root29-Aspire-A715-42G:~$ sudo cat /etc/splunkforwarder/etc/system/local/outputs.conf
[sudo] password for root29:
tcpout
tcpout::default-authn-group
tcpout::default-authn-group
server = 192.168.0.102:9999
tcpout::server://192.168.0.102:9999
root29@root29-Aspire-A715-42G:~$
```

Inputs.conf: - Configured log monitoring for authentication logs.



C. Network Connectivity Test (ping):-Verified network connectivity

between host and Splunk server.



```
root@root23-Azure-A715-426:~# ping 192.168.0.102
PING 192.168.0.102: 192.168.0.102: 36 bytes of data:
64 bytes from 192.168.0.102: icmp_seq=1 ttl=64 time=0.390 ms
64 bytes from 192.168.0.102: icmp_seq=2 ttl=64 time=0.295 ms
64 bytes from 192.168.0.102: icmp_seq=3 ttl=64 time=0.281 ms
64 bytes from 192.168.0.102: icmp_seq=4 ttl=64 time=0.322 ms
64 bytes from 192.168.0.102: icmp_seq=5 ttl=64 time=0.290 ms
64 bytes from 192.168.0.102: icmp_seq=6 ttl=64 time=0.283 ms
64 bytes from 192.168.0.102: icmp_seq=7 ttl=64 time=0.327 ms
64 bytes from 192.168.0.102: icmp_seq=8 ttl=64 time=0.362 ms
64 bytes from 192.168.0.102: icmp_seq=9 ttl=64 time=0.257 ms
64 bytes from 192.168.0.102: icmp_seq=10 ttl=64 time=0.305 ms
64 bytes from 192.168.0.102: icmp_seq=11 ttl=64 time=0.314 ms
64 bytes from 192.168.0.102: icmp_seq=12 ttl=64 time=0.294 ms
64 bytes from 192.168.0.102: icmp_seq=13 ttl=64 time=0.305 ms
64 bytes from 192.168.0.102: icmp_seq=14 ttl=64 time=0.407 ms
64 bytes from 192.168.0.102: icmp_seq=15 ttl=64 time=0.277 ms
64 bytes from 192.168.0.102: icmp_seq=16 ttl=64 time=0.306 ms
64 bytes from 192.168.0.102: icmp_seq=17 ttl=64 time=0.205 ms
64 bytes from 192.168.0.102: icmp_seq=18 ttl=64 time=0.208 ms
64 bytes from 192.168.0.102: icmp_seq=19 ttl=64 time=0.232 ms
64 bytes from 192.168.0.102: icmp_seq=20 ttl=64 time=0.208 ms
%
--- 192.168.0.102 ping statistics ---
20 packets transmitted, 20 received, 0% packet loss, time 1940ms
rtt min/avg/max/mdev = 0.302/0.325/0.514/0.093 ms
root@root23-Azure-A715-426:~#
```

4. Log Monitoring & Detection

In this step, logs forwarded from the host machine are analyzed in Splunk to detect suspicious activities and simulate real-world SOC operations.

4.1 Verifying Log Ingestion

To confirm that logs are successfully reaching the Splunk server, a basic search is performed:

Search Query:
index=*

This displays all incoming logs from the host system, confirming successful log ingestion.

4.2 Simulating an Attack

To test the detection capability, a brute-force login attempt is simulated on the host system using incorrect SSH login attempts.

Command:

```
ssh wronguser@localhost
```

Multiple failed login attempts are performed to generate suspicious activity logs.

4.3 Detecting Failed Login Attempts

The following search query is used in Splunk to detect failed login attempts:

Search Query:

```
index=* "Failed password"
```

This helps identify potential brute-force attacks by showing repeated login failures.

4.4 Creating an Alert

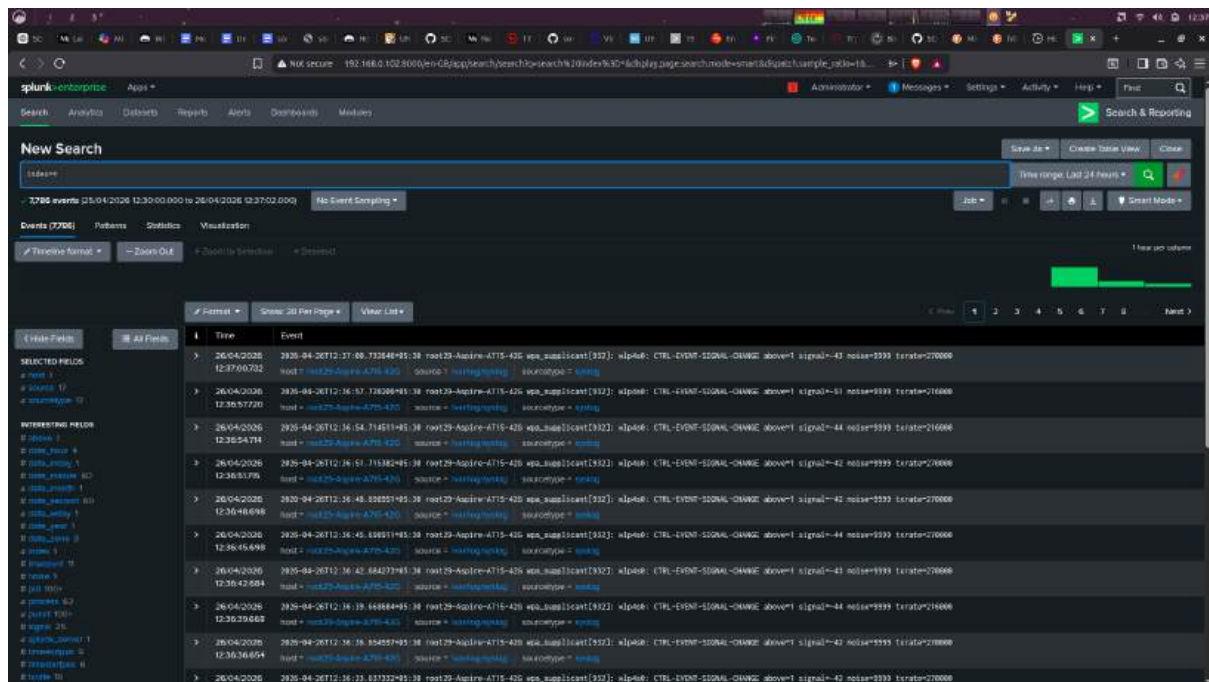
To automate detection, an alert is created in Splunk based on failed login attempts.

Steps:

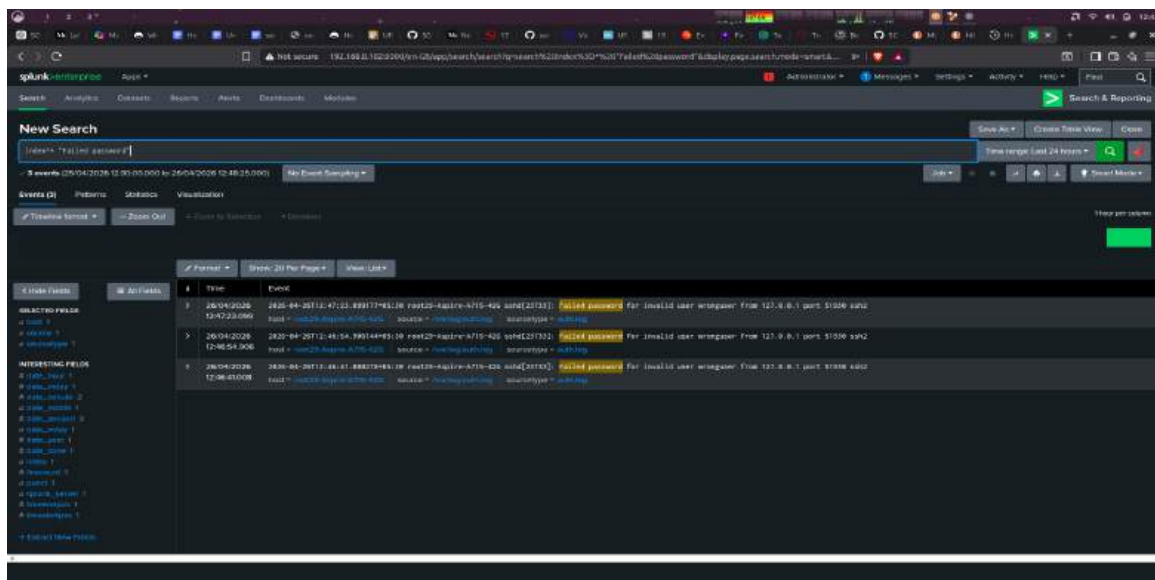
- Perform the search for failed passwords
- Click on “Save As” → Alert
- Set conditions (e.g., more than 5 failed attempts in 5 minutes)

This enables real-time alerting for suspicious login activity.

A. Logs in Splunk (index=*)



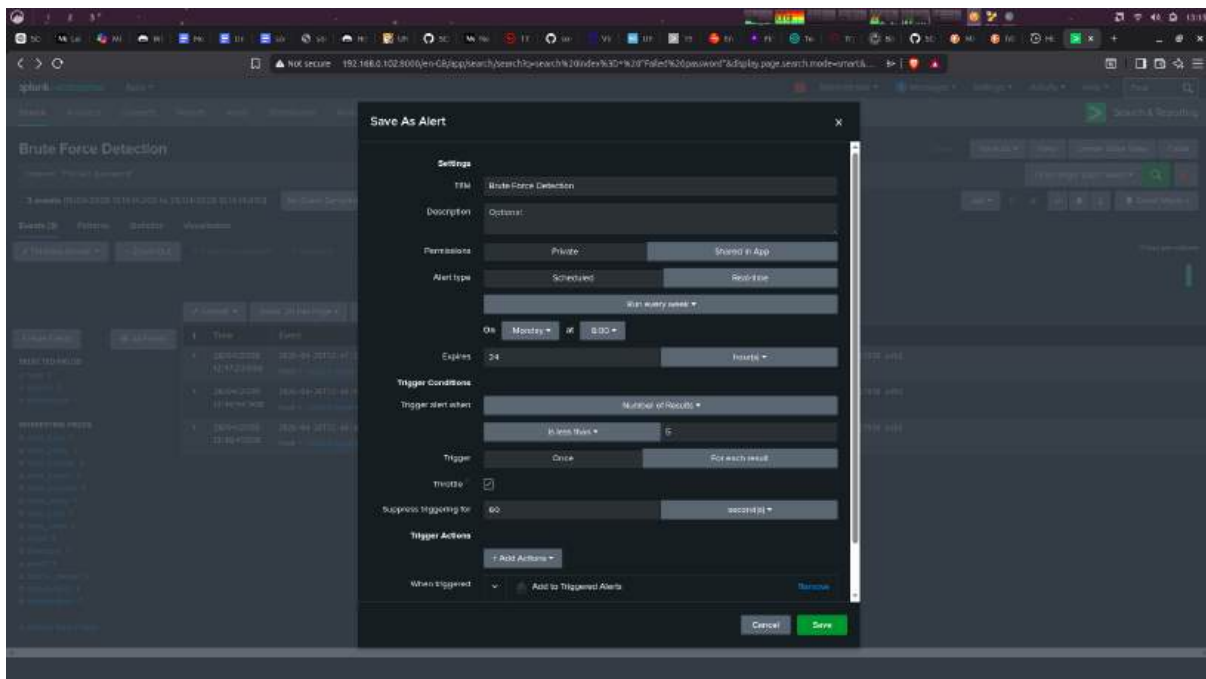
B. Failed Login Detection ("Failed password")



C. Attack Simulation (SSH brute force)

```
root29@root29-Aspire-A715-426:~$ ssh wronguser@localhost
The authenticity of host 'localhost (127.0.0.1)' can't be established.
ED25519 key fingerprint is SHA256:WA13VNHhZfSKmTnXkyLtkvWdCwTLWXSfghetuLnpmMs.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
wronguser@localhost's password:
Permission denied, please try again.
wronguser@localhost's password:
Permission denied, please try again.
wronguser@localhost's password:
wronguser@localhost: Permission denied (publickey,password).
root29@root29-Aspire-A715-426:~$
```

D. Alert Creation



5. Conclusion

This project demonstrates the successful setup of end-to-end Security Operations Center (SOC) lab using Splunk Enterprise and Universal Forwarder.

Through this lab, logs were collected from a Linux Mint host system and forwarded to a centralized Splunk server running on a virtual machine. The project also included real-time log monitoring, attack simulation, and detection of failed login attempts.

A brute-force attack scenario was simulated using multiple failed SSH login attempts, and the activity was successfully detected in Splunk. Additionally, an alert was created to automatically identify suspicious login behavior.

This project helped in understanding the practical workflow of a SOC analyst, including log collection, monitoring, detection, and alerting.

Future Improvements

- Creating dashboards for better visualization
- Simulating advanced attacks using penetration testing tools
- Integrating multiple log sources
- Implementing automated response mechanisms

Overall, this lab provides a strong foundation for real-world SOC operations and cybersecurity analysis.

6. Troubleshooting

During the setup of the SOC lab, several common issues can occur. This section provides step-by-step troubleshooting techniques to identify and resolve them.

6.1 Forwarder Not Connecting to Splunk

Problem:

The Universal Forwarder is unable to connect to the Splunk server.

What to Check:

- Verify Splunk server IP address

- Check if port 9997 is open and listening
- Ensure network connectivity between host and VM

Commands to Run:

- Check forwarder status:
sudo /opt/splunkforwarder/bin/splunk status
- Verify connection configuration:
cat /opt/splunkforwarder/etc/system/local/outputs.conf
- Test network connectivity:
ping <Splunk_Server_IP>
- Check port connection:
telnet <Splunk_Server_IP> 9997

Solution:

- Correct the IP address in outputs.conf
- Enable receiving on port 9997 in Splunk
- Fix VirtualBox network settings (use Bridged Adapter)
- Restart the forwarder after changes

6.2 Logs Not Appearing in Splunk

Problem:

Logs are not visible in the Splunk interface

What to Check:

- Ensure inputs.conf is properly configured
- Verify log file path exists
- Confirm forwarder is running

Commands to Run:

- Check monitored logs:
cat /opt/splunkforwarder/etc/system/local/inputs.conf
- Verify log file exists:
ls /var/log/auth.log
- Restart forwarder:
sudo /opt/splunkforwarder/bin/splunk restart
- Check Splunk logs:
index=_internal

Solution:

- Add correct log path in inputs.conf
- Ensure proper permissions on log files

- Restart forwarder after configuration
 - Wait a few seconds for logs to appear
-

6.3 Alert Not Triggering

Problem:

Alert is created but not getting triggered.

What to Check:

- Verify search query returns results
- Check alert condition and time range
- Ensure sufficient failed attempts are generated

Commands / Steps:

- Run search manually:
`index=* "Failed password"`
- Simulate attack:
`ssh wronguser@localhost`

Solution:

- Reduce threshold (e.g., >2 for testing)
 - Adjust time range (last 5 minutes)
 - Generate multiple failed login attempts
 - Ensure alert is enabled
-

This troubleshooting process helps in identifying and resolving common issues in a SOC lab environment, improving practical debugging skills required for real-world security operations.

7. Securing Log Transmission using SSL

To enhance security, SSL encryption was explored between the Universal Forwarder and the Splunk server to ensure secure log transmission over the network.

7.1 Certificate Generation

A self-signed SSL certificate was generated on the Splunk server (Ubuntu VM) using OpenSSL.

The certificate included the server IP address as the Common Name (CN) to support secure communication between the forwarder and the server.

7.2 Enabling SSL on Splunk Server

SSL configuration was attempted on port 9997 by modifying the `inputs.conf` file.

The Splunk server was configured to use the generated certificate for encrypted data receiving.

7.3 Configuring Forwarder for SSL

The Universal Forwarder on the Linux Mint system was configured to forward logs over SSL by updating the `outputs.conf` file and adding SSL-related parameters.

7.4 Challenges Faced

During implementation, several configuration issues were encountered, including certificate validation errors and service instability.

These issues affected the normal functioning of the Splunk server.

To maintain a stable logging environment, the SSL configuration was rolled back, and the setup was restored to a non-SSL configuration.

7.5 Learning Outcome

This process provided valuable hands-on experience in understanding SSL configuration, certificate management, and troubleshooting real-world issues related to secure log transmission in a SOC environment.

8. Interview Preparation

8.1 Project Explanation

This project demonstrates the setup of a end-to-end Security Operations Center (SOC) lab using Splunk Enterprise and Universal Forwarder.

A Linux Mint system was used as the log-generating machine, where the Universal Forwarder was installed to collect system logs.

A virtual machine running Ubuntu was used as the Splunk server to receive and analyze logs.

Log forwarding was configured using `outputs.conf` and `inputs.conf`, enabling real-time log ingestion.

To simulate a real-world attack scenario, multiple failed SSH login attempts were generated to mimic a brute-force attack. These events were successfully detected in Splunk using search queries, and an alert was created to identify suspicious activity.

This project provided hands-on experience in log collection, monitoring, detection, and alerting, which are core responsibilities of a SOC analyst.

8.2 Important Interview Questions & Answers

Q1: What is Splunk and why did you use it?

Answer:

Splunk is a SIEM tool used for collecting, analyzing, and monitoring machine data such as logs.

In this project, it was used to centralize logs from a host machine and detect suspicious activities like failed login attempts.

Q2: What is a Universal Forwarder?

Answer:

Universal Forwarder is a lightweight agent that collects logs from a system and forwards them to the Splunk server.

In this project, it was installed on Linux Mint to send authentication logs to the Splunk server.

Q3: How does log forwarding work in your setup?

Answer:

Log forwarding is configured using `outputs.conf` and `inputs.conf`.

The `outputs.conf` defines the destination Splunk server IP address and port, while `inputs.conf` specifies which logs to monitor.

The forwarder continuously sends logs to the Splunk server over port 9997.

Q4: How did you detect a brute-force attack?

Answer:

Multiple failed SSH login attempts were simulated, and the Splunk query `index=* "Failed password"` was used to identify repeated login failures.

This pattern indicates a potential brute-force attack.

Q5: What challenges did you face in this project?

Answer:

Challenges included forwarder connectivity issues, log ingestion problems, and SSL configuration difficulties.

These were resolved by verifying configuration files, checking network connectivity, and troubleshooting system logs.

This provided practical experience in real-world SOC troubleshooting.

9. Best Practices & Maintenance

9.1 Regular Maintenance

Daily Checks

- Check Splunk service status:

```
sudo /opt/splunk/bin/splunk status
```

- Verify Universal Forwarder status:

```
sudo /opt/splunkforwarder/bin/splunk status
```

- Run basic search in Splunk Web:

index=*

- Verify that logs are being received
-

Weekly Checks

- Check disk usage:

df -h

- Verify forwarder connection:

sudo /opt/splunkforwarder/bin/splunk list forward-server

- Check internal Splunk logs:

index=_internal

- Ensure alerts are triggering correctly
-

Monthly Checks

- Clean or archive old logs
- Restart Splunk for health check:

sudo /opt/splunk/bin/splunk restart

- Apply system updates:

sudo apt update && sudo apt upgrade

- Take backup of configuration files
-

9.2 Log Retention Policy

Log Retention Duration

- Small lab: 7–30 days
 - Production: 90+ days
-

Disk Space Management

- Monitor disk usage:

```
df -h
```

- Configure index size limits in `indexes.conf`
-

Old Log Deletion

- Manual deletion:

```
sudo rm -rf /opt/splunk/var/lib/splunk/*
```

- Prefer using Splunk retention policies
-

9.3 Security Best Practices

Access Control

- Use strong passwords
 - Change default credentials
 - Implement role-based access control
-

Forwarder Security

- Connect forwarder only to trusted server
 - Keep port 9997 open, close unnecessary ports
 - Implement SSL in future
-

Dashboard Security

- Disable public access
 - Hide sensitive data
 - Restrict access based on user roles
-

9.4 Monitoring Tips

Performance Monitoring

- Check Splunk internal logs:

`index=_internal`

- Monitor CPU, memory, and indexing performance
-

Alert Tuning

- Adjust thresholds (keep low for testing)
 - Optimize time ranges (5 min, 10 min, etc.)
 - Avoid excessive alerts
-

Reducing False Positives

- Use specific queries
 - Exclude known IP addresses
 - Refine alert conditions
-

9.5 Future Improvements

Enhancements

- Add multiple log sources (Apache, Syslog)
 - Simulate attacks using Kali Linux
 - Centralize logs from multiple systems
-

Advanced Features

- Create Splunk dashboards
 - Implement correlation rules
 - Integrate threat intelligence
-

Pro-Level Upgrades

- Proper SSL implementation
- Develop SIEM use cases

- Add automated response (SOAR basics)

10. Glossary

Term	Description
SOC	Security Operations Center – team that monitors and responds to security incidents
SIEM	Security Information and Event Management system used for log analysis
Splunk	SIEM tool used for collecting and analyzing logs
Universal Forwarder	Lightweight agent used to send logs to Splunk server
Log	Record of system events and activities
Brute Force Attack	Repeated login attempts to guess credentials
SSH	Secure Shell protocol for remote login
Index	Storage location in Splunk where logs are stored
Alert	Notification triggered based on defined conditions
inputs.conf	File used to define which logs to collect
outputs.conf	File used to define where logs should be sent